

Aspects of Multivariate Analysis

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Content adapted from:

Johnson, R. A., & Wichern, D. W. (2007). Applied Multivariate Statistical Analysis (6th ed).

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Multivariate Nature of Data and Research

Most studies collected multivariate data

- Collect multiple explanatory (predictor) variables
- Collect multiple outcome (response) variables
- Collect variable(s) at multiple occasions

Most (interesting) research questions involve multiple variables

- Understand similarities/differences between variables
- Understand associations between variables
- Understand structure inherent to variables

Objectives of Multivariate Methods

1. Data reduction or simplification
 - Find simple representation to improve interpretation
2. Sorting and grouping
 - Classify or cluster “similar” observations
3. Investigation of dependence
 - Explore nature and degree of associations
4. Prediction
 - Explaining one or more outcome variables
5. Hypothesis testing
 - Multivariate tests of location, correlation, regression, etc.

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Data Reduction or Simplification

Summarize multivariate data via **composite score**

- Combine cancer patient outcome data to radiotherapy response
- Combine international athlete data to make performance metric

Simplification for **interpretation**:

- Visualize high-altitude multispectral image data in 2D
- Genetic selection for improved crop yield and protein
- Determine how professional mediators resolve disputes

Sorting and Grouping

Clustering: places similar observations into groups

- Cluster jobs based on computer usage (for future planning)
- Cluster tax returns into suspicious (audit) and okay (no audit)

Classification: distinguishes between pre-existing groups

- Discriminate alcoholics using psychophysiological variables
- Discriminate MS visual pathology using visual stimuli response

Investigation of Dependence

Find **optimal combinations** of variables for producing outcomes:

- Identify factors responsible for client success with consultants
- Identify pulp fiber properties that produce high quality paper

Explore **general associations** between variables

- Investigate factors that lead some firms to be innovators
- Explore relationships between risk-taking and performance

Prediction

Predict a **continuous outcome**

- Predict college success from test scores and HS performance
- Predict depositional environments from sediment distributions

Predict a **nominal outcome**

- Classify insolvent insurers from accounting and financial variables
- Classify cancer tumor type from cDNA microarray data

Hypothesis Construction and Testing

Test multivariate differences between two groups

- Test differences in pollution levels on weekends vs weekdays
- Test competing sociological theories for American occupations

Test nature of multivariate associations

- Determine if nature of instructions affects perceived risk in task
- Determine firm innovation patterns in industrialize countries

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The Data Matrix

The **data matrix** refers to the array of numbers

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ x_{31} & x_{32} & \cdots & x_{3p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$

where x_{ij} is the j -th variable collected from the i -th item (subject).

- items/subjects are rows
- variables are columns

$\mathbf{X}$ is a data matrix of order $n \times p$ ($\#$ items by $\#$ variables).

Collection of Column Vectors

We can view a data matrix as a collection of **column vectors**:

$$\mathbf{X} = \begin{bmatrix} | & | & \cdots & | \\ \mathbf{x}_1 & \mathbf{x}_2 & \cdots & \mathbf{x}_p \\ | & | & & | \end{bmatrix}$$

where $\mathbf{x}_j$ is the j -th column of $\mathbf{X}$ for $j \in \{1, \dots, p\}$.

The $n \times 1$ vector $\mathbf{x}_j$ gives the j -th variable's scores for the n items.

Collection of Row Vectors

We can view a data matrix as a collection of **row vectors**:

$$\mathbf{X} = \begin{bmatrix} \text{---} & \mathbf{x}_1^\top & \text{---} \\ \text{---} & \mathbf{x}_2^\top & \text{---} \\ & \vdots & \\ \text{---} & \mathbf{x}_n^\top & \text{---} \end{bmatrix}$$

where $\mathbf{x}_i^\top$ is the i -th row of $\mathbf{X}$ for $i \in \{1, \dots, n\}$.

The $1 \times p$ vector $\mathbf{x}_i^\top$ gives the i -th item's scores for the p variables.

Example: University Book Store Receipts

Suppose that we collect $n = 4$ receipts from the University book store, and we record $p = 2$ variables from each receipt:

- **Variable 1:** the amount of money spent (dollars) 42, 52, 48, 58
- **Variable 2:** the number of books purchased (count) 4, 5, 4, 3

Variable 1: $x_{11} = 42$ $x_{21} = 52$ $x_{31} = 48$ $x_{41} = 58$

Variable 2: $x_{12} = 4$ $x_{22} = 5$ $x_{32} = 4$ $x_{42} = 3$

Collecting these numbers produces a data matrix of order 4×2

$$\mathbf{X} = \begin{bmatrix} 42 & 4 \\ 52 & 5 \\ 48 & 4 \\ 58 & 3 \end{bmatrix}$$

Calculating Variable (Column) Means

The sample mean of the j -th variable is given by

$$\begin{aligned}\bar{x}_j &= \frac{1}{n} \sum_{i=1}^n x_{ij} \\ &= n^{-1} \mathbf{1}_n^\top \mathbf{x}_j\end{aligned}$$

where

- $\mathbf{1}_n$ denotes an $n \times 1$ vector of ones
- $\mathbf{x}_j$ denotes the j -th column of $\mathbf{X}$

Calculating Item (Row) Means

The sample mean of the i -th item is given by

$$\begin{aligned}\bar{x}_i &= \frac{1}{p} \sum_{j=1}^p x_{ij} \\ &= p^{-1} \mathbf{x}_i^\top \mathbf{1}_p\end{aligned}$$

where

- $\mathbf{1}_p$ denotes an $p \times 1$ vector of ones
- $\mathbf{x}_i^\top$ denotes the i -th row of $\mathbf{X}$

Example: Mean Vector Calculation

The column means have the form

$$\bar{x}_1 = \frac{1}{4} \sum_{i=1}^4 x_{i1} = \frac{1}{4}(42 + 52 + 48 + 58) = 50$$

$$\bar{x}_2 = \frac{1}{4} \sum_{i=1}^4 x_{i2} = \frac{1}{4}(4 + 5 + 4 + 3) = 4$$

so the mean vector has the form

$$\bar{\mathbf{x}} = \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \end{bmatrix} = \begin{bmatrix} 50 \\ 4 \end{bmatrix}$$

The Variation and Covariation of Data

The **covariance matrix** refers to the symmetric array of numbers

$$\mathbf{S} = \begin{bmatrix} s_1^2 & s_{12} & s_{13} & \cdots & s_{1p} \\ s_{21} & s_2^2 & s_{23} & \cdots & s_{2p} \\ s_{31} & s_{32} & s_3^2 & \cdots & s_{3p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ s_{p1} & s_{p2} & s_{p3} & \cdots & s_p^2 \end{bmatrix}$$

where

- $s_j^2 = (1/n) \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2$ is the **variance** of the j -th variable
- $s_{jk} = (1/n) \sum_{i=1}^n (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)$ is the **covariance** between the j -th and k -th variables
- $\bar{x}_j = (1/n) \sum_{i=1}^n x_{ij}$ is the mean of the j -th variable

Covariance Matrix from Data Matrix

We can calculate the covariance matrix such as

$$\mathbf{S} = \frac{1}{n} \mathbf{X}_c^\top \mathbf{X}_c$$

where $\mathbf{X}_c = \mathbf{X} - \mathbf{1}_n \bar{\mathbf{x}}^\top = \mathbf{C}\mathbf{X}$ with

- $\bar{\mathbf{x}}^\top = (\bar{x}_1, \dots, \bar{x}_p)$ denoting the vector of variable means
- $\mathbf{C} = \mathbf{I}_n - n^{-1} \mathbf{1}_n \mathbf{1}_n^\top$ denoting a centering matrix

Note that the centered matrix $\mathbf{X}_c$ has the form

$$\mathbf{X}_c = \begin{bmatrix} x_{11} - \bar{x}_1 & x_{12} - \bar{x}_2 & \cdots & x_{1p} - \bar{x}_p \\ x_{21} - \bar{x}_1 & x_{22} - \bar{x}_2 & \cdots & x_{2p} - \bar{x}_p \\ x_{31} - \bar{x}_1 & x_{32} - \bar{x}_2 & \cdots & x_{3p} - \bar{x}_p \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} - \bar{x}_1 & x_{n2} - \bar{x}_2 & \cdots & x_{np} - \bar{x}_p \end{bmatrix}$$

Example: Variance and Covariance Calculation

The variance and covariance terms have the forms

$$\begin{aligned}s_1^2 &= \frac{1}{4} \sum_{i=1}^4 (x_{i1} - \bar{x}_1)^2 = \frac{1}{4} \sum_{i=1}^4 x_{i1}^2 - \bar{x}_1^2 \\ &= \frac{1}{4} (42^2 + 52^2 + 48^2 + 58^2) - 50^2 = 34\end{aligned}$$

$$\begin{aligned}s_2^2 &= \frac{1}{4} \sum_{i=1}^4 (x_{i2} - \bar{x}_2)^2 = \frac{1}{4} \sum_{i=1}^4 x_{i2}^2 - \bar{x}_2^2 \\ &= \frac{1}{4} (4^2 + 5^2 + 4^2 + 3^2) - 4^2 = 1/2\end{aligned}$$

$$\begin{aligned}s_{12} &= \frac{1}{4} \sum_{i=1}^4 (x_{i1} - \bar{x}_1)(x_{i2} - \bar{x}_2) = \frac{1}{4} \sum_{i=1}^4 x_{i1}x_{i2} - \bar{x}_1\bar{x}_2 \\ &= \frac{1}{4} (42(4) + 52(5) + 48(4) + 58(3)) - 50(4) = -3/2\end{aligned}$$

The Correlation of Data

The **correlation matrix** refers to the symmetric array of numbers

$$\mathbf{R} = \begin{bmatrix} 1 & r_{12} & r_{13} & \cdots & r_{1p} \\ r_{21} & 1 & r_{23} & \cdots & r_{2p} \\ r_{31} & r_{32} & 1 & \cdots & r_{3p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ r_{p1} & r_{p2} & r_{p3} & \cdots & 1 \end{bmatrix}$$

where

$$r_{jk} = \frac{s_{jk}}{s_j s_k} = \frac{\sum_{i=1}^n (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)}{\sqrt{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2} \sqrt{\sum_{i=1}^n (x_{ik} - \bar{x}_k)^2}}$$

is the Pearson correlation coefficient between variables $\mathbf{x}_j$ and $\mathbf{x}_k$.

Correlation Matrix from Data Matrix

We can calculate the correlation matrix such as

$$\mathbf{R} = \frac{1}{n} \mathbf{X}_s^\top \mathbf{X}_s$$

where $\mathbf{X}_s = \mathbf{C}\mathbf{X}\mathbf{D}^{-1}$ with

- $\mathbf{C} = \mathbf{I}_n - n^{-1} \mathbf{1}_n \mathbf{1}_n^\top$ denoting a centering matrix
- $\mathbf{D} = \text{diag}(s_1, \dots, s_p)$ denoting a diagonal scaling matrix

Note that the standardized matrix $\mathbf{X}_s$ has the form

$$\mathbf{X}_s = \begin{bmatrix} (x_{11} - \bar{x}_1)/s_1 & (x_{12} - \bar{x}_2)/s_2 & \cdots & (x_{1p} - \bar{x}_p)/s_p \\ (x_{21} - \bar{x}_1)/s_1 & (x_{22} - \bar{x}_2)/s_2 & \cdots & (x_{2p} - \bar{x}_p)/s_p \\ (x_{31} - \bar{x}_1)/s_1 & (x_{32} - \bar{x}_2)/s_2 & \cdots & (x_{3p} - \bar{x}_p)/s_p \\ \vdots & \vdots & \ddots & \vdots \\ (x_{n1} - \bar{x}_1)/s_1 & (x_{n2} - \bar{x}_2)/s_2 & \cdots & (x_{np} - \bar{x}_p)/s_p \end{bmatrix}$$

Example: Covariance and Correlation Matrix

The covariance matrix has the form

$$\mathbf{S} = \begin{bmatrix} 34 & -3/2 \\ -3/2 & 1/2 \end{bmatrix}$$

The correlation matrix has the form

$$\mathbf{R} = \begin{bmatrix} 1 & -0.3638 \\ -0.3638 & 1 \end{bmatrix}$$

where the correlation coefficient is calculated as

$$r_{12} = \frac{s_{12}}{s_1 s_2} = \frac{-3/2}{\sqrt{34}\sqrt{1/2}} = -0.3638$$

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Scatter Plot with Dot Diagrams

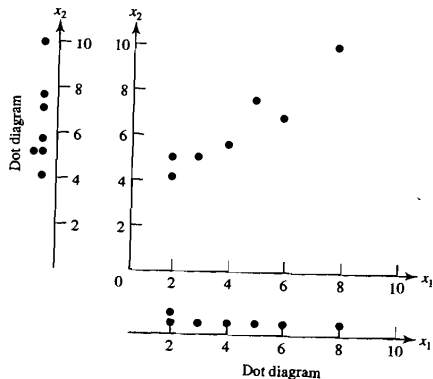


Figure 1.1 A scatter plot and marginal dot diagrams.

Figure 1: Figure 1.1 from Johnson and Wichern (2007).

Different Scatter Plot with Same Dot Diagrams

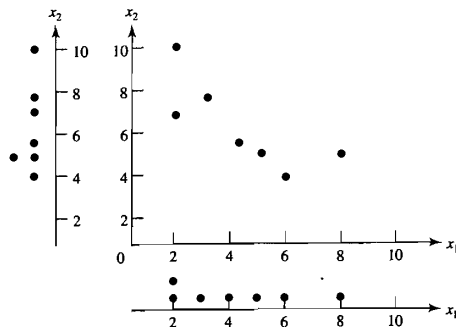


Figure 1.2 Scatter plot and dot diagrams for rearranged data.

Figure 2: Figure 1.2 from Johnson and Wichern (2007).

Paneled Scatter Plots with Box Plots

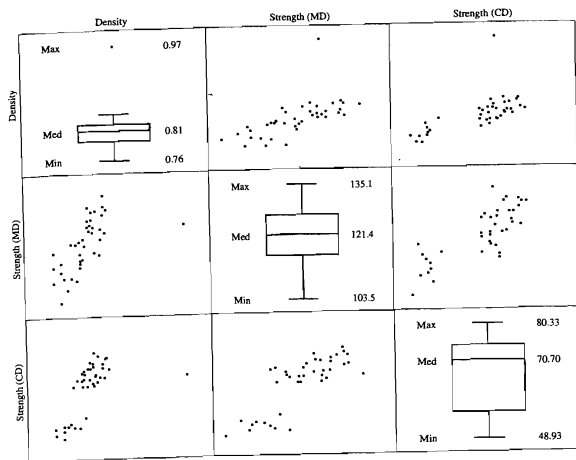


Figure 1.5 Scatter plots and boxplots of paper-quality data from Table 1.2.

Figure 3: Figure 1.5 from Johnson and Wichern (2007).

3D Scatter Plots

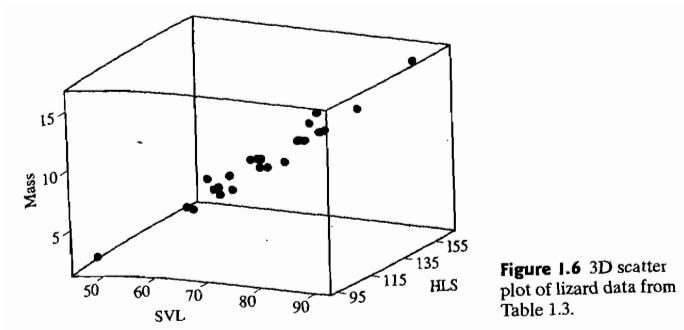


Figure 4: Figure 1.6 from Johnson and Wichern (2007).

3D Scatter Plots with Grouping (= 4D Scatter Plot)

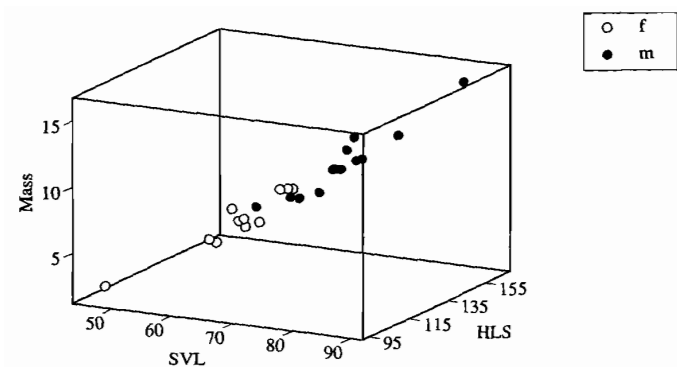


Figure 1.8 3D scatter plot of male and female lizards.

Figure 5: Figure 1.8 from Johnson and Wichern (2007).

Perspective Matters (Rotate!)

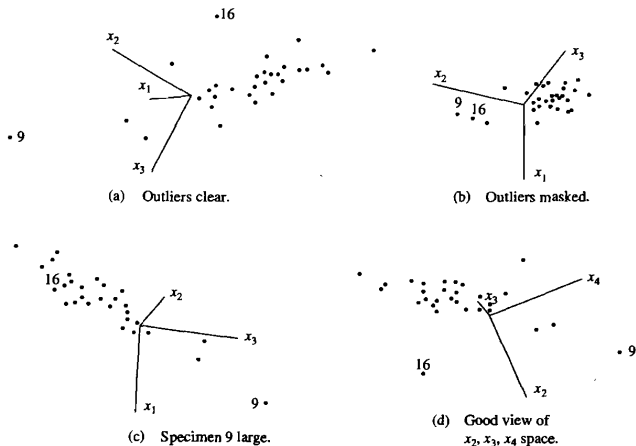


Figure 1.12 Three-dimensional perspectives for the lumber stiffness data.

Figure 6: Figure 1.12 from Johnson and Wichern (2007).

Combined Growth Curves

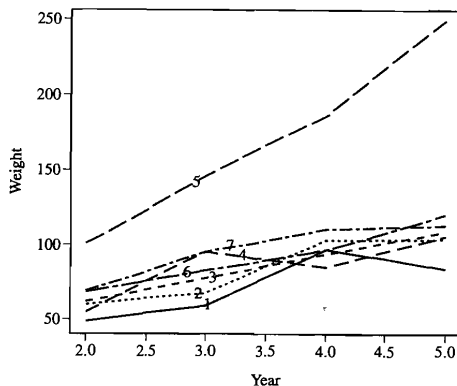


Figure 1.13 Combined growth curves for weight for seven female grizzly bears.

Figure 7: Figure 1.13 from Johnson and Wichern (2007).

Individual Growth Curves

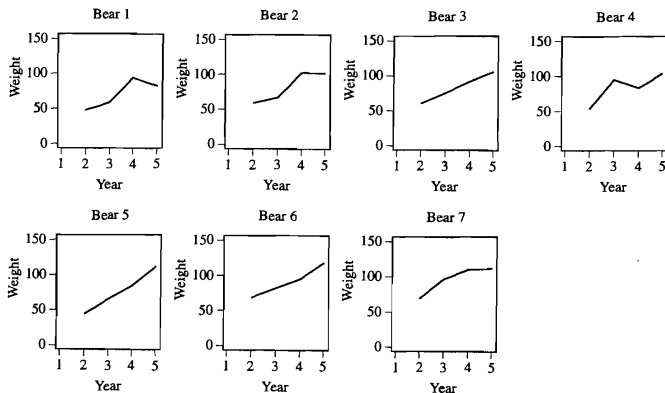


Figure 1.14 Individual growth curves for weight for female grizzly bears.

Figure 8: Figure 1.14 from Johnson and Wichern (2007).

Star/Spider/Radar Charts for $p = 8$ variables

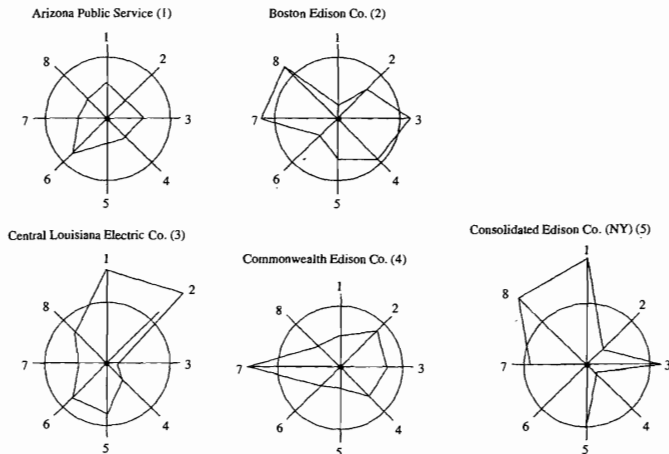


Figure 1.16 Stars for the first five public utilities.

Figure 9: Figure 1.16 from Johnson and Wichern (2007).

Chernoff Faces Example

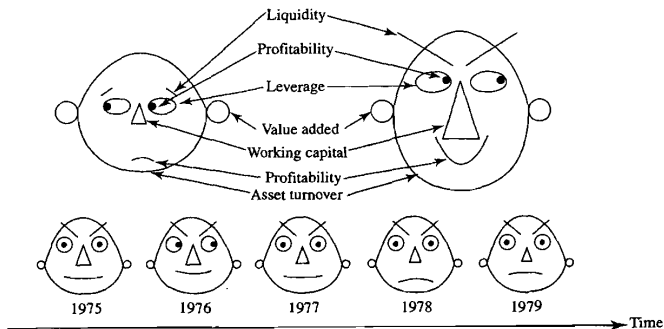


Figure 1.18 Chernoff faces over time.

Figure 10: Figure 1.18 from Johnson and Wichern (2007).

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Length as Distance to Origin

Let $P = (x_1, \dots, x_p)$ denote a point in $p \geq 2$ dimensional space

Let $O = (0, \dots, 0)$ denote the **origin** (p dimensional vector of zeros)

Pythagorean theorem reveals that **length** is distance to origin

$$d(O, P) = \sqrt{x_1^2 + x_2^2 + \dots + x_p^2}$$

where $d(\cdot, \cdot)$ denotes the Euclidean distance (defined on next slide).

Euclidean Distance Between Points

Let $P = (x_1, \dots, x_p)$ denote a point in $p \geq 2$ dimensional space

Let $Q = (y_1, \dots, y_p)$ denote a point in $p \geq 2$ dimensional space

The Euclidean distance between P and Q is given by

$$\begin{aligned} d(P, Q) &= \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \\ &= \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_p - y_p)^2} \end{aligned}$$

Heterogeneity of Variances

Euclidean distance only makes sense if units (i.e., variances) of the variables are comparable...

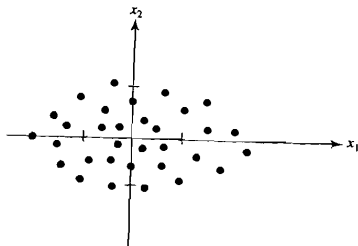


Figure 1.20 A scatter plot with greater variability in the x_1 direction than in the x_2 direction.

Figure 11: Figure 1.20 from Johnson and Wichern (2007).

Statistical Distance

The **statistical distance** between P and Q is given by

$$\begin{aligned} d(P, Q) &= \sqrt{\sum_{i=1}^n \left(\frac{x_i}{s_i} - \frac{y_i}{s_i} \right)^2} \\ &= \sqrt{\frac{(x_1 - y_1)^2}{s_1} + \frac{(x_2 - y_2)^2}{s_2} + \dots + \frac{(x_p - y_p)^2}{s_p}} \end{aligned}$$

which is the Euclidean distance calculated from the standardized points

$$P_* = (x_1/s_1, \dots, x_p/s_p)$$

$$Q_* = (y_1/s_1, \dots, y_p/s_p)$$

where s_j^2 denotes the variance of the j -th variable.

Ellipse of Constant Statistical Distance

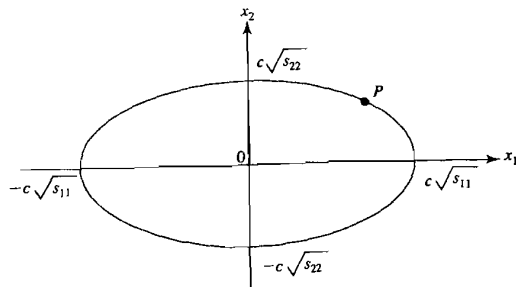


Figure 1.21 The ellipse of constant statistical distance
 $d^2(O, P) = x_1^2/s_{11} + x_2^2/s_{22} = c^2$.

Figure 12: Figure 1.21 from Johnson and Wichern (2007).

Orthogonal Rotation Changes Coordinate System

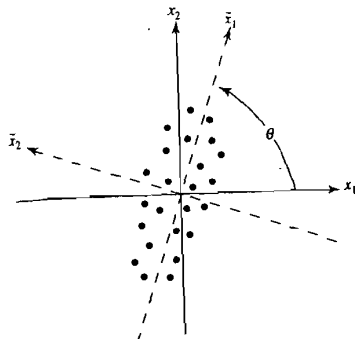


Figure 1.23 A scatter plot for positively correlated measurements and a rotated coordinate system.

Figure 13: Figure 1.23 from Johnson and Wichern (2007).

$$\tilde{x}_1 = x_1 \cos(\theta) + x_2 \sin(\theta)$$

$$\tilde{x}_2 = -x_1 \sin(\theta) + x_2 \cos(\theta)$$

Ellipse of Constant Distance in Rotated Coordinates

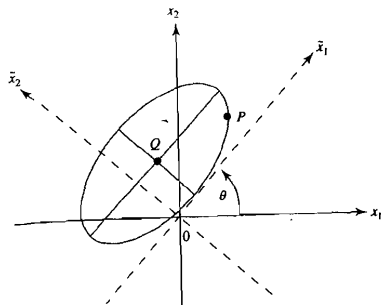


Figure 1.24 Ellipse of points a constant distance from the point Q .

Figure 14: Figure 1.24 from Johnson and Wichern (2007).

Properties of Distances

$$d(P, Q) = d(Q, P) \quad \text{symmetry}$$

$$d(P, Q) > 0 \text{ if } P \neq Q \quad \text{non-negativity}$$

$$d(P, Q) = 0 \text{ if } P = Q \quad \text{identity}$$

$$d(P, Q) \leq d(P, R) + d(R, Q) \quad \text{triangle inequality}$$